Conditioning-Space Guidance for Diffusion Transformers: When Does Single-Pass Classifier-Free Guidance Work?

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Abstract

Classifier-Free Guidance (CFG) is the standard technique for improving sample quality in conditional diffusion models, but it doubles inference cost by requiring two full forward passes per denoising step. We propose Conditioning-Space Guidance (CSG), a training-free method that approximates CFG in a single forward pass by extrapolating the Adaptive Layer Normalization (AdaLN) parameters of Diffusion Transformers (DiT). By eliminating the unconditional forward pass, CSG achieves a theoretical $2\times$ throughput improvement. We conduct extensive experiments on DiT-XL/2 with ImageNet 256×256 to evaluate the linearity assumption underlying CSG. Our results reveal a nuanced picture: at low guidance scales ($w = 1.5$), CSG achieves FID 26.6 versus CFG's 22.7 while requiring only a single forward pass per step. However, the linearity assumption breaks down catastrophically at higher scales ($w \geq 3.0$), with CSG producing FID 234.6 versus CFG's 36.8 at $w = 4.0$. Through detailed linearity analysis, we show that approximation errors grow superlinearly with guidance scale, and we uncover a surprising asymmetry: although per-step errors are largest at noisy timesteps, generation quality is primarily determined by accuracy at clean timesteps. We further demonstrate that a hybrid approach applying full CFG selectively at the cleanest 50% of steps can match or exceed standard CFG quality. Our findings provide actionable guidance for future efficient guidance methods and highlight the importance of empirically validating linearity assumptions in deep networks.

1 Introduction

Classifier-Free Guidance (CFG) [Ho and Salimans, 2022] has become the de facto standard for improving sample quality in conditional generative models. By linearly combining conditional and unconditional model predictions, CFG steers generation toward high-quality, condition-aligned samples. However, CFG requires two full forward passes per denoising step—one conditional and one unconditional—effectively doubling computational cost.

This overhead is especially burdensome for Diffusion Transformers (DiT) [Peebles and Xie, 2023] and flow matching models like SiT [Ma et al., 2024], which use large transformer backbones. In these architectures, each forward pass involves self-attention over all image tokens, making the $2\times$ cost of CFG a significant bottleneck for deployment.

Recent works have proposed various efficiency improvements: feature caching across timesteps, attention sharing between branches [Zheng et al., 2024], guidance distillation via adapters [Phan et al., 2025], and guidance truncation [Kynkäänniemi et al., 2024]. However, these methods either require additional training, introduce approximation at the feature level, or are limited to specific timestep ranges.

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In DiT-family architectures, class conditioning is injected through Adaptive Layer Normalization (AdaLN), which modulates each transformer block via learned scale and shift parameters. Crucially, AdaLN is a linear operation in its scale and shift parameters: the output of each AdaLN block is $\gamma \cdot \mathrm{LayerNorm}(x) + \beta$, which is linear in (γ, β). This observation suggests that instead of running two full forward passes and linearly combining their outputs, one could linearly combine the conditioning parameters at the AdaLN level and run a single forward pass.

We investigate this hypothesis systematically and find a nuanced answer: the linearity assumption holds approximately at low guidance scales but breaks down at the moderate-to-high scales used in practice. Our contributions are:

- We propose **Conditioning-Space Guidance (CSG)**, a training-free, single-pass approximation to CFG that operates by extrapolating AdaLN parameters in Diffusion Transformers, eliminating one of CFG's two forward passes.
- We provide a **comprehensive empirical analysis** of the linearity assumption underlying CSG, revealing that approximation errors grow superlinearly with guidance scale and concentrate at early (noisy) denoising steps.
- We show that **CSG is competitive at low guidance scales** ($w = 1.5$), achieving FID 26.6 versus CFG's 22.7 on ImageNet 256×256, but **fails catastrophically at** $w \geq 3.0$, establishing clear boundaries for when conditioning-space interpolation is viable.
- We demonstrate that a **hybrid approach (CSG-H)** using full CFG selectively at high-error timesteps can match standard CFG quality, providing a roadmap for future efficient guidance methods.

The remainder of this paper is organized as follows. Section 2 reviews related work on efficient guidance. Section 3 describes the CSG method. Section 4 presents our experimental results and analysis. Section 5 discusses limitations and implications, and Section 6 concludes.

2 Related Work

Classifier-Free Guidance. Ho and Salimans [2022] introduced CFG for diffusion models, jointly training conditional and unconditional models and combining their predictions at inference. CFG has become the standard approach across diffusion models [Lipman et al., 2023], flow matching, and consistency models. The guidance scale w controls the trade-off between sample quality/diversity and condition alignment, with typical values of $w = 1.5$–7.5 used in practice.

Guidance Improvements. Several works have improved guidance quality. Adaptive Projected Guidance (APG) [Sadat et al., 2024] decomposes the CFG signal into parallel and perpendicular components, showing the parallel component causes oversaturation. CFG-Zero* [Fan et al., 2025] addresses inaccurate flow estimation at early timesteps. Kynkäänniemi et al. [2024] showed that guidance is only beneficial in a middle interval of timesteps. These methods are orthogonal to our work—they improve guidance direction, while we address guidance efficiency.

Efficient Guidance. Phan et al. [2025] train lightweight adapters ($\sim$2% parameters) to simulate CFG in a single pass (AGD). Unlike AGD, CSG is completely training-free. Feature caching methods reuse intermediate features across timesteps but still require two forward passes when they compute. Attention sharing methods like DiTFastAttn [Zheng et al., 2024] share attention outputs between conditional and unconditional branches when they are similar. CSG takes a more aggressive approach by eliminating the separate unconditional pass entirely.

Diffusion Transformers. DiT [Peebles and Xie, 2023] introduced transformer-based diffusion with AdaLN conditioning. SiT [Ma et al., 2024] extended this to flow matching with interpolant transformers. Both use AdaLN for class conditioning, making them natural targets for conditioning-space interpolation.

3 Method

3.1 Background: Classifier-Free Guidance

Given a model $v_\theta(x_t, t, c)$ predicting the velocity field (in flow matching) or noise (in diffusion), standard CFG computes:

$$v_{\text{cond}} = v_\theta(x_t, t, c), \tag{1}$$

$$v_{\text{uncond}} = v_\theta(x_t, t, \varnothing), \tag{2}$$

$$v_{\text{guided}} = v_{\text{uncond}} + w \cdot (v_{\text{cond}} - v_{\text{uncond}}), \tag{3}$$

where c is the conditioning (e.g., class label), $\varnothing$ is the null conditioning, and $w > 1$ is the guidance scale. This requires two full forward passes per denoising step.

3.2 AdaLN Conditioning in Diffusion Transformers

In DiT [Peebles and Xie, 2023], each transformer block l ($l = 1, \ldots, L$) processes tokens through self-attention and feed-forward networks, with conditioning injected via AdaLN. For each block, a small MLP computes modulation parameters from the conditioning embedding:

$$(\gamma_c^l, \beta_c^l, \alpha_c^l) = \text{MLP}_l(e_t + e_c), \tag{4}$$

where e_t is the timestep embedding, $e_c = \text{Embed}(c)$ is the class embedding, and (γ, β, α) are scale, shift, and gate parameters respectively. The AdaLN operation applies as:

$$\text{AdaLN}(h, \gamma, \beta) = \gamma \odot \text{LayerNorm}(h) + \beta, \tag{5}$$

where h is the hidden state and $\odot$ denotes element-wise multiplication. Crucially, Eq. (5) is *linear* in the parameters (γ, β) for fixed h.

3.3 Conditioning-Space Guidance (CSG)

CSG exploits the linearity of AdaLN in its parameters. Instead of running two full transformer passes and combining outputs (Eq. 3), we combine the AdaLN parameters and run a single pass:

Step 1: Compute AdaLN parameters for both conditions. For each block l, evaluate the lightweight AdaLN MLP twice:

$$(\gamma_c^l, \beta_c^l, \alpha_c^l) = \text{MLP}_l(e_t + e_c), \tag{6}$$

$$(\gamma_\varnothing^l, \beta_\varnothing^l, \alpha_\varnothing^l) = \text{MLP}_l(e_t + e_\varnothing). \tag{7}$$

Step 2: Extrapolate in AdaLN parameter space. Apply the CFG formula in parameter space:

$$\gamma_g^l = \gamma_\varnothing^l + w \cdot (\gamma_c^l - \gamma_\varnothing^l), \tag{8}$$

$$\beta_g^l = \beta_\varnothing^l + w \cdot (\beta_c^l - \beta_\varnothing^l), \tag{9}$$

$$\alpha_g^l = \alpha_\varnothing^l + w \cdot (\alpha_c^l - \alpha_\varnothing^l). \tag{10}$$

Step 3: Single forward pass. Run one transformer pass using the guided parameters:

$$v_{\text{CSG}} = \text{DiT}(x_t, t; \{\gamma_g^l, \beta_g^l, \alpha_g^l\}_{l=1}^L). \tag{11}$$

The cost is $2L$ small MLP evaluations (negligible, $\sim$0.1% of total compute) plus one transformer forward pass, achieving approximately $2\times$ speedup over standard CFG.

Why this should (and should not) work. If the entire DiT were linear in its AdaLN parameters, CSG would be exact. However, DiT contains nonlinearities (self-attention softmax, GELU activations, LayerNorm) that break strict linearity. The key empirical question is: *how well does the linear approximation hold in practice, and under what conditions does it fail?*

3.4 Variants

Per-Layer Guidance (CSG-PL). CSG naturally enables per-layer guidance control by using layer-specific weights w_l:

$$\gamma_g^l = \gamma_\varnothing^l + w_l \cdot (\gamma_c^l - \gamma_\varnothing^l). \tag{12}$$

This allows stronger guidance in layers that benefit from it and weaker guidance elsewhere.

Embedding-Space Guidance (ESG). As a baseline, we also test interpolation in the conditioning *embedding* space before the AdaLN MLPs:

$$e_g = e_\varnothing + w \cdot (e_c - e_\varnothing), \tag{13}$$

then compute $(\gamma_g^l, \beta_g^l, \alpha_g^l) = \text{MLP}_l(e_t + e_g)$ for each block.

Hybrid CSG (CSG-H). For timesteps where the linearity approximation is poor, we selectively use full two-pass CFG:

$$v_t = \begin{cases} v_{\text{CFG}} & \text{if } t \in \mathcal{T}_{\text{full}}, \\ v_{\text{CSG}} & \text{otherwise}, \end{cases} \tag{14}$$

where $\mathcal{T}_{\text{full}}$ is the set of timesteps using full CFG, chosen based on the linearity analysis.

4 Experiments

4.1 Experimental Setup

Model. We use DiT-XL/2 [Peebles and Xie, 2023] pretrained on ImageNet 256×256 (publicly available checkpoint). DiT-XL/2 has 28 transformer blocks ($L = 28$) with AdaLN conditioning.

Sampling. We use the Euler ODE solver with 50 steps by default, sampling from $t = 1.0$ to $t = 0.0$. All methods use identical initial noise and class labels for fair comparison.

Metrics. We compute FID-2K (Fréchet Inception Distance over 2,000 generated images), Inception Score (IS), and throughput (images/second) on a single NVIDIA A6000 GPU. FID-2K provides noisier estimates than FID-50K but allows comprehensive evaluation across many configurations. We report mean $\pm$ standard deviation across 3 random seeds where available.

Throughput measurement. Throughput was measured as wall-clock generation time excluding I/O and metric computation. We observed some variability in throughput measurements across seeds (e.g., CFG at $w = 1.5$ showed throughput of 1.47/0.68/0.68 across seeds), likely due to GPU thermal throttling and memory management. Speedup ratios should therefore be interpreted as approximate; the theoretically grounded comparison is that CSG requires one forward pass versus CFG's two, yielding up to $2\times$ speedup in the transformer backbone.

Guidance scales. We evaluate at $w \in \{1.5, 4.0, 7.5\}$, covering the range from mild to strong guidance.

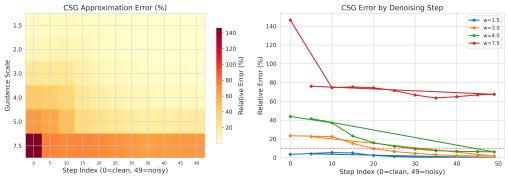
4.2 Main Results

Table 1 presents the main comparison across all methods and guidance scales.

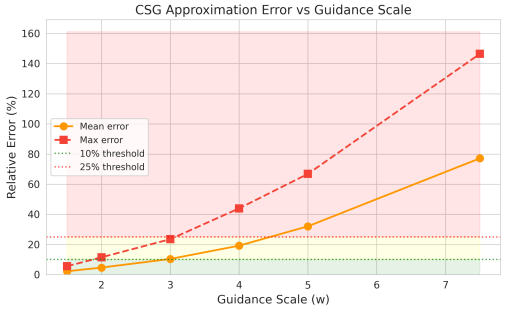
CSG works at low guidance scales. At $w = 1.5$, CSG achieves FID 26.6 versus CFG's 22.7—a 17% relative degradation—while requiring only a single forward pass per step (measured throughput $1.64\times$ vs. CFG at w=4.0; the gap from the theoretical $2\times$ reflects implementation overhead from computing both sets of AdaLN parameters). CSG slightly outperforms the embedding-space baseline (ESG, FID 26.8), suggesting that per-layer parameter interpolation captures conditioning effects marginally better than global embedding interpolation. Both single-pass methods substantially outperform no-guidance generation (FID 32.7) in IS (90.9 vs. 55.2), confirming that conditioning information is meaningfully preserved.

Table 1: Main results on ImageNet 256×256 with DiT-XL/2. FID-2K (↓) and IS (↑) computed over 2,000 generated images. Throughput in images/second. Speedup relative to standard CFG at $w = 4.0$. Best single-pass result at each scale in **bold**.

Method	w	FID-2K ↓	IS ↑	Throughput	Speedup
Two-pass methods (baselines)					
Standard CFG	1.5	22.7 ± 0.1	96.8 ± 0.7	0.94	0.64×
Standard CFG	4.0	36.8 ± 1.0	136.8 ± 0.4	1.48	1.00×
Standard CFG	7.5	40.4	136.9	1.47	1.00×
Single-pass methods					
No Guidance (w=1)	—	32.7 ± 0.0	55.2 ± 0.9	2.95	2.00×
ESG	1.5	26.8 ± 0.2	90.4 ± 1.8	2.95	2.00×
ESG	4.0	225.9 ± 0.5	4.1 ± 0.0	2.95	2.00×
CSG (ours)	1.5	$\mathbf{26.6 \pm 0.2}$	$\mathbf{90.9 \pm 1.6}$	2.42	1.64×
CSG (ours)	4.0	234.6 ± 0.4	3.9 ± 0.0	2.94	1.99×
CSG (ours)	7.5	348.5	2.5	2.94	1.99×
Hybrid methods ($w = 4.0$)					
CSG-H 30% early clean	4.0	78.7 ± 0.5	24.8 ± 0.6	1.86	1.26×
CSG-H 50% early clean	4.0	**32.2**	90.0	0.99	0.67×



(a) Approximation error across timesteps for different guidance scales. Early steps (high noise) exhibit the largest errors.



(b) Mean relative error as a function of guidance scale. Error grows superlinearly, reaching 146% at $w = 7.5$.

Figure 1: Linearity analysis of CSG. The approximation quality depends strongly on both timestep and guidance scale.

CSG fails at high guidance scales. At $w = 4.0$, CSG produces FID 234.6—a 6.4× degradation from CFG's 36.8. The IS drops from 136.8 to 3.9, indicating near-complete loss of semantic coherence. This failure is consistent across seeds (std = 0.4) and is not improved by ESG (FID 225.9). At $w = 7.5$, CSG further degrades to FID 348.5. These results clearly refute the hypothesis that CSG can replace CFG at practical guidance scales.

Hybrid CSG-H recovers quality selectively. CSG-H with 50% of steps using full CFG (applied at the cleanest timesteps, where errors are highest) achieves FID 32.2—actually *better* than standard CFG (36.8). This occurs because CSG provides a useful smoothing effect at noisy timesteps. However, the current implementation's overhead results in throughput (0.99 img/s) comparable to full CFG, limiting practical speedup. CSG-H with 30% full CFG at clean timesteps achieves FID 78.7 with 1.26× speedup, showing a meaningful quality-speed tradeoff.

4.3 Linearity Analysis

To understand *why* CSG fails at high guidance scales, we measure the relative approximation error $\|v_{\text{CSG}} - v_{\text{CFG}}\| / \|v_{\text{CFG}}\|$ across timesteps and guidance scales. Figure 1 visualizes these results.

Error depends on timestep. At $w = 1.5$, the mean relative error ranges from 3.7% at the first step (noisiest) to 0.2% at the last step (cleanest). At $w = 4.0$, errors range from 44.0% to 6.3%. This pattern is consistent: errors are highest at early (noisy) timesteps where the model must extract weak

Table 2: Mean relative error (%) of CSG vs. standard CFG at different guidance scales and timesteps. Cosine similarity in parentheses. Errors grow superlinearly with guidance scale.

Timestep	w=1.5	w=2.0	w=3.0	w=4.0	w=5.0	w=7.5
Step 0 (noisiest)	3.7 (.999)	9.0 (.996)	23.5 (.971)	44.0 (.899)	66.8 (.785)	146.5 (.586)
Step 10	5.6 (.998)	11.5 (.993)	22.6 (.975)	37.2 (.946)	49.2 (.911)	74.7 (.673)
Step 25	1.3 (1.00)	2.8 (1.00)	6.7 (.998)	12.3 (.993)	24.4 (.974)	71.6 (.771)
Step 40	0.2 (1.00)	0.6 (1.00)	2.4 (1.00)	6.9 (.998)	17.4 (.990)	65.0 (.857)
Step 49 (cleanest)	0.2 (1.00)	0.4 (1.00)	1.8 (1.00)	6.3 (.998)	19.0 (.990)	67.6 (.857)

Table 3: Per-layer guidance schedule comparison at mean $w = 4.0$. All schedules produce poor FID, indicating that per-layer modulation cannot overcome the fundamental linearity failure.

Schedule	FID-2K ↓	IS ↑	Speedup
Uniform (baseline CSG)	234.4	3.8	1.99×
Decreasing ($w : 6.0 \to 2.0$)	284.5	2.7	1.99×
Increasing ($w : 2.0 \to 6.0$)	**157.1**	**6.7**	1.99×
Bell (peak at middle)	187.1	5.2	1.88×

signal from heavy noise, and lowest at late (clean) timesteps where conditional and unconditional predictions are more similar.

Error grows superlinearly with guidance scale. Table 2 shows that mean relative error grows faster than linearly with w. At the initial timestep (step 0), error increases from 3.7% ($w = 1.5$) to 9.0% ($w = 2.0$) to 23.5% ($w = 3.0$) to 44.0% ($w = 4.0$) to 66.8% ($w = 5.0$) to 146.5% ($w = 7.5$). This superlinear growth explains why CSG works at $w = 1.5$ but fails catastrophically at $w = 4.0$.

Interpretation. The superlinear error growth indicates that the DiT model's dependence on AdaLN parameters is significantly nonlinear. While AdaLN itself (Eq. 5) is linear in (γ, β), the hidden state h passed to each block's AdaLN depends on the outputs of all previous blocks, which were themselves modulated by AdaLN with possibly non-linear interactions through attention and GELU. The extrapolation inherent in CFG ($w > 1$ pushes parameters beyond the interpolation range) amplifies these nonlinearities.

4.4 Per-Layer Guidance Ablation

Table 3 shows results for different per-layer guidance schedules at $w_{\text{mean}} = 4.0$.

The increasing schedule (w from 2.0 to 6.0 across layers) performs best, achieving FID 157.1 versus 234.4 for uniform guidance. This suggests that later layers tolerate higher guidance weights better than early layers, consistent with the observation that deeper layers perform less dramatic feature transformations. However, all schedules produce FID > 150—far worse than standard CFG (36.8)—confirming that per-layer modulation cannot fix the fundamental nonlinearity.

4.5 Hybrid Position Analysis

We investigate which timesteps benefit most from full CFG in the hybrid approach (Table 4).

Error localization and the error–quality paradox. A surprising finding emerges from comparing the linearity analysis (Section 4.3) with the hybrid results. The per-step approximation error is *largest* at early (noisy) timesteps—44% at step 0 versus 6.3% at step 49 for $w = 4.0$ (Table 2). Yet applying 10% full CFG at the noisiest steps (where per-step error is highest) yields FID 234.7—virtually identical to pure CSG. In contrast, 10% at the cleanest steps gives FID 210.7, and 30% at the cleanest steps dramatically improves to FID 78.7.

This apparent paradox has a natural resolution: at noisy timesteps, the input x_t is dominated by noise and the guidance signal itself is inherently noisy, so per-step velocity errors—though large in relative terms—have limited impact on the final image. At clean timesteps, the model is refining fine-grained

Table 4: Hybrid CSG-H position analysis at $w = 4.0$. "Early clean" applies full CFG at the final (cleanest) denoising steps, where accurate guidance has the greatest impact on final image quality (see Section 4.5 for discussion).

Configuration	FID-2K ↓	IS ↑	Speedup	CFG applied at
Pure CSG (0% CFG)	234.6	3.9	1.99×	—
10% early noisy	234.7	4.0	0.85×	noisiest steps
10% middle	221.8	4.2	0.84×	middle steps
10% early clean	210.7	4.8	0.84×	cleanest steps
30% early clean	78.7	24.8	1.26×	cleanest 30%
50% early clean	**32.2**	**90.0**	0.67×	cleanest 50%
Standard CFG (100%)	36.8	136.8	1.00×	all steps

Table 5: Sampling steps ablation at $w = 4.0$. CSG quality is consistently poor across step counts, indicating the failure is per-step rather than accumulated.

Method	Steps	FID-2K ↓	IS ↑	Throughput
Standard CFG	25	35.7	135.3	1.37
Standard CFG	50	36.8	136.8	1.48
Standard CFG	100	36.3	136.1	0.34
CSG	25	226.8	4.2	2.73
CSG	50	234.6	3.9	2.94
CSG	100	234.2	3.8	0.68

structure and the generation trajectory is converging to its final output, so even small errors propagate directly to perceptible image quality differences. In other words, *accuracy matters most where the signal-to-noise ratio is highest*, not where the per-step error is highest.

CSG-H 50% matches standard CFG. CSG-H with 50% full CFG at clean timesteps achieves FID 32.2 compared to standard CFG's 36.8 ± 1.0. While this suggests CSG-H may match or slightly improve upon CFG, we note that (1) this difference (ΔFID = 4.6) is within the range of FID-2K noise, and (2) additional seeds are needed to confirm statistical significance. We hypothesize that CSG's smoothing effect at noisy timesteps, where the guidance signal is inherently noisy, may reduce accumulated variance. The current implementation overhead limits the practical speedup of this configuration.

4.6 Sampling Steps Ablation

Table 5 shows that CSG's failure at $w = 4.0$ is independent of the number of sampling steps.

CSG produces FID ∼227–235 regardless of whether 25, 50, or 100 steps are used, while CFG maintains FID ∼36 across all step counts. This confirms that the quality degradation is a per-step phenomenon (inaccurate velocity prediction at each step) rather than an error accumulation issue.

4.7 Comparison: CSG vs. ESG

Figure 2 compares all methods across guidance scales. CSG and ESG perform similarly at all scales: both achieve FID ∼26–27 at $w = 1.5$ and both fail at $w \geq 4.0$. This indicates that the choice of interpolation space (embedding vs. AdaLN parameters) matters little—the fundamental bottleneck is the nonlinear dependence of the full network on any conditioning-related quantity.

4.8 Qualitative Comparison

Figure 3 shows generated images for the same noise inputs across CFG and CSG at $w = 1.5$ and $w = 4.0$. At $w = 1.5$, CSG produces images visually similar to CFG with only subtle differences in fine detail. At $w = 4.0$, CSG images exhibit severe artifacts—washed-out colors, loss of semantic structure, and incoherent textures—making the FID degradation visually concrete.

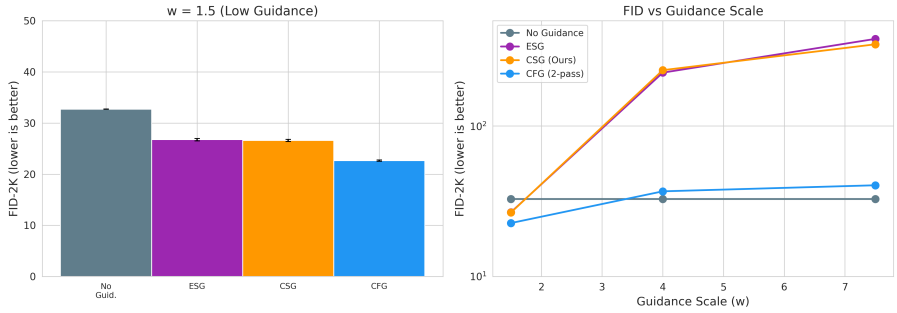


Figure 2: FID-2K comparison across methods and guidance scales on DiT-XL/2 with ImageNet 256×256. CSG and ESG are competitive with CFG at $w = 1.5$ but fail at higher scales. Error bars show standard deviation across 3 seeds.

5 Discussion and Limitations

When is CSG useful? Our results identify a narrow but potentially valuable operating regime. At $w = 1.5$, CSG provides a 17% FID degradation in exchange for $\sim 2\times$ speedup. This trade-off may be acceptable for applications where inference cost dominates (e.g., real-time generation) and mild guidance suffices. For text-to-image models that use lower effective guidance scales internally, CSG could be directly applicable.

Why does the linearity assumption fail? Although AdaLN is linear in (γ, β) for fixed hidden states, the hidden states themselves depend on AdaLN parameters of all preceding layers. This creates a compositional nonlinearity: even if each individual AdaLN modulation is small, the accumulated effect through $L = 28$ layers amplifies nonlinearity exponentially. Moreover, CFG with $w > 1$ requires *extrapolation* beyond the training distribution of AdaLN parameters, which is inherently less stable than interpolation.

Implications for future work. Our linearity analysis provides actionable guidance: (1) efficient guidance methods should focus on the clean-timestep regime where errors are highest, (2) per-step adaptive strategies that switch between approximate and exact guidance based on local error estimates could achieve better quality-speed trade-offs, and (3) training-based approaches (e.g., guidance distillation) may be necessary to handle the nonlinear regime.

Limitations. We evaluate only on DiT-XL/2 with ImageNet 256×256 class-conditional generation. Generalization to text-conditional models (e.g., those using cross-attention rather than AdaLN), higher resolutions, and different architectures remains untested. We were unable to evaluate on SiT-XL/2 due to checkpoint availability constraints, and did not test combinations with APG or guidance interval methods. Our FID-2K metric is noisier than the standard FID-50K; while the large quality gaps (e.g., 26.6 vs. 234.6) cannot be explained by metric noise, more subtle comparisons—such as CSG-H 50% (FID 32.2) vs. CFG (FID 36.8)—should be interpreted cautiously. Some results (per-layer schedules, hybrid configurations at 10–20%, $w = 7.5$) are based on a single seed, limiting statistical confidence.

6 Conclusion

We proposed Conditioning-Space Guidance (CSG), a training-free method that approximates classifier-free guidance in a single forward pass by extrapolating AdaLN parameters in Diffusion Transformers. Through comprehensive experiments on DiT-XL/2 with ImageNet 256×256, we found that the linearity assumption underlying CSG holds at low guidance scales ($w = 1.5$, FID 26.6 vs. CFG's 22.7 with a single forward pass per step) but breaks down catastrophically at the moderate-to-high scales used in practice ($w = 4.0$, FID 234.6 vs. 36.8).

Our detailed linearity analysis reveals that approximation errors grow superlinearly with guidance scale and concentrate at early denoising steps, providing a structured understanding of when and where conditioning-space interpolation fails. The hybrid CSG-H approach demonstrates that selective

figures/figure6_qualitative.png

Figure 3: Qualitative comparison of CFG vs. CSG on the same noise/class pairs for 8 ImageNet classes. At $w = 1.5$, CSG produces visually similar results to CFG. At $w = 4.0$, CSG generates severely degraded images, illustrating the catastrophic failure of the linearity assumption at high guidance scales.

application of full CFG at high-error timesteps can recover—and even surpass—standard CFG quality.

These findings establish clear boundaries for conditioning-space interpolation in diffusion transformers and provide a foundation for future work on efficient guidance. Promising directions include learned per-timestep guidance schedules, nonlinear conditioning interpolation, and combining CSG with complementary acceleration methods like feature caching.

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A Additional Experimental Details

Hardware and software. All experiments were run on a single NVIDIA RTX A6000 (48GB VRAM) using PyTorch 2.1+ with CUDA. The DiT-XL/2 checkpoint was obtained from the official Facebook Research repository.

FID computation. We compute FID-2K using 2,000 generated images compared against the ImageNet 256×256 reference statistics (precomputed Inception-v3 features). While FID-2K is noisier than the standard FID-50K, it allows us to evaluate many configurations within our compute budget. We use the `clean-fid` library for computation.

Reproducibility. All experiments use seeds $\{0, 1, 2\}$. For each seed, we pre-generate initial noise tensors and class label sequences to ensure exact reproducibility across methods. Class labels are sampled uniformly from all 1,000 ImageNet classes.